FORENSIC TRIAGE REPORT — FORENSIGUARD

Timestamp (UTC): 2026-10-21T12:15:45.490705+00:00 | **Tool:** ForensiGuard v1.0.0

1. Evidence Provenance & Cryptographic Identifiers

Attribute	Value
Filename	src_0000_obfuscated.jpg
File Size	313 bytes
SHA-256	bdf588cee79cbc7a7e779d0c5d36b105...
MD5	78f2d44af0224dc78282690b60a71073
Declared Ext / MIME	.jpg / application/pdf
Signature Check	MISMATCH (ANOMALY)

2. Probabilistic Model Inferences & Technique Attribution

Verdict	TAMPERED / MANIPULATED
Manipulation Probability	90.93%
Attributed Technique	FILE_TYPE_OBFUSCATION
Confidence Score	87.81%
Metadata XGBoost Score	0.37%
Image Residual CNN Score	N/A (Non-image / Signature Mismatch)
Clean-Baseline Anomaly	0.3497

3. Forensic Narrative & Explanations

• [OBFUSCATION CONFIRMED] Binary magic bytes (application/pdf) directly contradict the declared file extension ('.jpg').

• Time interval between modification and metadata change (mtime - ctime) (observed: -1.17, SHAP: -4.843) supported benign classification.

• Time interval between access and modification (atime - mtime) (observed: 182.17, SHAP: -0.499) supported benign classification.

• Isolation Forest deviation from clean filesystem baseline (observed: 0.35, SHAP: -0.025) supported benign classification.

• Time interval between access and metadata change (atime - ctime) (observed: 181.00, SHAP: -0.025) supported benign classification.

4. Legal Disclaimer & Evidentiary Scope

ForensiGuard provides probabilistic forensic triage indicators based on machine-learning models. These inferences do NOT constitute proof of guilt or conclusive evidentiary admissibility under Daubert/Frye. Independent manual examination of filesystem journal logs, file slack, and carving is mandatory.